INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

CUP, WATER CANTEEN, W/WIRE HANDLE, CORROSION-RESISTING STEEL

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers one type and size of corrosion-resisting steel canteen cup with a folding wire handle (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

NATICK LABORATORY ARMY RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-9-284 Cup, Canteen; Assembly and Details

(Copies of this document are available from the procuring activity.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

AMERICAN WELDING SOCIETY

AWS C1.4M/C1.4 - Specification for Resistance Welding of Carbon and Low-Allow Steels
AWS D17.1/D17.1M - Specification for Fusion Welding for Aerospace Applications

(Copies of these documents are available online at https://www.aws.org.)

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A967/A967M - Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM E18 - Standard Test Methods for Rockwell Hardness of Metallic Materials

(Copies of these documents are available online at https://www.astm.org.)

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SAE INTERNATIONAL (SAE)

AMS QQ-S-763 - Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings

(Copies of this document are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Toxic chemicals, hazardous substances, and ozone-depleting chemicals. The use of toxic chemicals, hazardous substances, or ozone-depleting chemicals shall be avoided, if feasible. The desired performance requirements should be specified rather than the specific chemical or substance. If a toxic chemical, hazardous substance, or ozone-depleting chemical shall be specified, it shall be listed as a key word in section 6. The Environmental Protection Agency maintains an online list of toxic chemicals and hazardous substances at https://www.epa.gov/epcra/consolidated-list-lists that should be consulted.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials and components. The materials and components shall be as specified herein and on Drawing 2-9-284. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document.

3.4.1 Steel, corrosion-resisting.

3.4.1.1 Sheet and strip. Corrosion-resisting steel sheet and strip for cup, (body and hinge), shall be class 302 or 304 of ASTM A240/A240M.

3.4.1.2 Wire. Corrosion-resisting steel wire shall conform to condition B, class 302 or 304, form round, finish cold of AMS QQ-S-763.

3.5 Design and construction. The design and construction of the canteen cup shall conform to Drawing 2-9-284 and the requirements specified herein (see 6.4).

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3.5.1 Body. The body of the cup shall be fabricated of the corrosion-resisting steel specified in 3.4.1.1 and shall be of seamless construction conforming to the dimensions specified on Drawing 2-9-284. The top edge shall be free of sharp edges, cracks, burrs, slivers, or rough surfaces. After forming, the vertical walls of the body shall have a minimum hardness of 46 on the 30N Rockwell scale when tested as specified in 4.5.5.

3.5.2 Handle. The handle shall be fabricated of corrosion-resisting steel specified in 3.4.1.2 and shall be welded in accordance with AWS D17.1/D17.1M or AWS C1.4M/C1.4. The resistance "T" welds of the handle shall withstand a minimum total force of 235 pounds when tested as specified in 4.5.4. The handle shall conform to the design, dimension, and detail requirements of Drawing 2-9-284. The handle shall fit snugly against the cup within 1/8-inch when folded and the right handle shall rest and lock into the left handle when in use.

3.5.3 Hinge. The hinge shall be fabricated of corrosion-resisting steel specified in 3.4.1.1 and shall conform to the design, dimension, and details specified on Drawing 2-9-284. The hinge shall be attached by spot welds or projection welds in accordance with AWS C1.4M/C1.4 and shall be located on the cup as shown on Drawing 2-9-284. The welds shall be free of splatter, burn holes, cracks or fractures, and discoloration and shall be tightly joined when tested as specified in 4.5.1, 4.5.2, or 4.5.3, as applicable. The hinge shall be a close fit to the contour of the body and there shall be no gaps between edge of hinge and body greater than 1/64-inch.

3.6 Finish. The canteen cup shall be given a uniform frosted lusterless finish approximating a commercial 2D sheet finish and passivated in accordance with AMS QQ-S-763 or ASTM A967/A967M. There shall be no evidence of rust, staining, pitting, or spotting when tested as specified in 4.5.6.

3.7 Marking. The letters US, the year of manufacture, and the manufacturer's name or trademark shall be stamped on the hinge as shown on Drawing 2-9-284.

3.8 Workmanship. The finished canteen cup shall be clean and free of any dent or being bent out of shape, stain, ripple, fracture, die or draw mark. The handles and hinge shall also be free of any sharp edge, burr, or sliver. Indentation on the inside of cup due to welding of the hinge is acceptable providing the indentation is smooth and free from sharp edges, weld splatter, burr, crack, or burn through area, and does not exceed 0.005 inch in depth.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the appropriate paragraph below.

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4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4.

4.4 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. In-process examination shall be made to establish conformance to the welding on wire handles requirements of Drawing 2-9-284. Whenever nonconformance is noted, correction shall be made to the applicable process and all items processed.

4.4.2.2 In-process spot or projection weld testing. The spot or projection weld sample peel test shall be conducted on two test specimens per welding machine rather than sample cups. The spot or projection welding testing shall be performed as specified in 4.5.1 at the start of each day's shifts production. Failure of test shall require correction of the welding process.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in Table I. The sample unit shall be one cup. The lot size shall be expressed in units of cups.

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TABLE I. End item visual defects.

Examine	Defect	Classification	
		Major	Minor
Finish	Not a uniform frosted, lusterless finish approximating a commercial 2D sheet finish		201
	Evidence of rust, staining, pitting, or spotting (slight water spotting, easily removed by washing shall not be considered a defect)	101	
Construction and Workmanship (applicable to all components and assemblies)	Seam, lap, laminations or crease, or metal ripple	102	
	Any component missing	103	
	Component dented		202
	Component fractured	104	
	Component bent out of shape		203
	Rim of cup contains sharp edge, burr, crack, sliver or rough spot	105	
	Evidence of die or draw mark		204
	Handle does not fold around cup within 1/8 inch as specified, or right handle does not nest into left handle	106	
	Hinge plate is not close fit against body, i.e. contains gap along edge greater than 1/64 inch (use feeler gage)		205
	Handle binds or requires force in opening or folding		206
	Handle or hinge contains sharp edge, burr, or sliver	107	
Welds	Contains crack or fracture	108	
	Not number specified	109	
	Not in location shown on drawing		207
	Contains burn hole, splatter, or discoloration	110	
	Evidence of poor fusion	111	
	Any missing weld	112	
	Weld on handle not neatly performed; e.g., contains rough or sharp projection or bead		208
	Indentation in weld area on inside of cup (Indentation on inside of the cup due to welding of the hinge is acceptable providing the indentation is smooth and free from sharp edges, weld splatter, burr, crack, burn through area, and does not exceed 0.005 inch in depth)		209
Marking	Missing, illegible or incomplete	113	

4.4.4 End item dimensional examination. The end items shall be examined for the dimensions annotated with an asterisk on Drawing 2-9-284. Any dimension not within the specified tolerance shall be classified as a defect. The sample unit shall be one cup. The lot size shall be expressed in units of cups.

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4.4.5 End item testing. The end items shall be tested as specified in 4.5.2 or 4.5.3, 4.5.4, 4.5.5, and 4.5.6. The sample unit shall be one cup. The lot size shall be expressed in units of cups.

4.5 Methods of inspection.

4.5.1 Spot or projection weld sample peel test. Each spot, or projection weld test specimen shall be of the type and thickness of material used in the cup body and hinge. Spot or projection weld each section as shown on Drawing 2-9-284 and then place in a vise or jig. Bend the parent metal piece and peel the hinge piece with pliers or another suitable tool. Inability of any spot weld to pull a 5/32-inch (minimum) hole in either piece of material shall constitute failure of this test. Failures of the projection welds or lack of fusion under the projection weld shall constitute failure of the test. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion.

4.5.2 Spot weld cup peel test. Place the cup in a jig or vise and peel the hinge from the cup. Inability of each weld to leave a 5/32-inch (minimum) hole in either the cup or hinge shall constitute a defect.

4.5.3 Projection weld cup peel test. Separate the weldment of the cup and hinge by driving a chisel between welded components in an unwelded area or by peeling the hinge back against the weld until failure occurs. Failure outside the periphery of the weld area shall be considered evidence that the welds are satisfactory. Lack of fusion under the projection weld shall constitute a defect. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion.

4.5.4 Test for resistance "T" welds of the handle. Place the cup in a jig or vise and exert a force against both "T" welds of the handle. Inability of any weld to withstand the minimum force specified in 3.5.2 shall constitute a defect.

4.5.5 Rockwell hardness test. Test the Rockwell hardness of the vertical walls of the body of the cup in accordance with ASTM E18. Take four Rockwell hardness readings on the vertical walls each 90 degrees apart around the cup and approximately half-way between the top and the bottom of the cup. Any sample cup having one or more readings less than 46 on the Rockwell 30N scale as specified in 3.5.1 shall be classified as a defect.

4.5.6 Finish test. Expose the complete assembled cup to 95-100 percent relative humidity at 72° ± 2 °F for not less than 16 hours to determine compliance with rust, staining, pitting, and spotting requirements of 3.6. Inability of the cup to conform to requirements of 3.6 shall constitute a defect.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory

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Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The corrosion-resisting steel cups are for use with canteen, water, plastic, MIL-C-43103. The cup is used for drinking hot and cold liquids, soups, and stews in the field.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity
c. National stock number.
d. Specific issue of individual documents referenced (see 2.2).
e. When first article is required (see 3.1, 4.3).
f. Name and address of the first article inspection facility; and the name and the address of the Government activity responsible for conducting the first article inspection program.
g. Inspection conditions.
h. Conformance inspection and acceptable quality limits.
i. Packaging required (see 5.1).

6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Hydroforming. Hydroforming has been found to produce acceptable cups.

6.5 Information requests. For access to information such as patterns, drawings, standard shade samples of cloth, samples, etc. go to DLA Troop Support's website for their Specification/Pattern/Drawing Request form, https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Complete the request form and then submit. Requests to use equivalent materials and/or components or to make changes to the pattern should be sent to the contracting officer for approval by the military.

6.6 Subject term (key word) listing.

Folding handle
Stainless steel
Mess kit

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6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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